

Prioritizing Potential Aquaculture

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Background

Considering that important role of marine aquaculture has in the global food supply, this analyses aim to determine which Exclusive Economic Zones (EEZ) on the West Coast of the US are best suited to developing marine aquaculture for several species of oysters and a species of commercial Atlantic Seabob Shrimp.

Data

- **Sea Surface Temperature:** Annual sea surface temperature (SST) from the years 2008 to 2012. The dataset can be found here: [NOAA's 5km Daily Global Satellite Sea Surface Temperature Anomaly v3.1](#).
- **Bathymetry:** The dataset can be found here: [General Bathymetric Chart of the Oceans \(GEBCO\)](#).³
- **Exclusive Economic Zones:** Maritime boundaries using Exclusive Economic Zones off of the west coast of US. The dataset can be found here: [Marineregions.orgs](#)

Load Packages

```
# Load libraries
library(here)
library(sf)
library(tidyverse)
library(tmap)
library(terra)
library(kableExtra)
```

Import data

Import the tree datasets: Sea Surface Temperature, Bathymetry and the Exclusive Economic Zones

```
# Read in raster sea surface temperature data
sea_temp_2008 <- rast(here("data", "data", "average_annual_sst_2008.tif"))
sea_temp_2009 <- rast(here("data", "data", "average_annual_sst_2009.tif"))
sea_temp_2010 <- rast(here("data", "data", "average_annual_sst_2010.tif"))
sea_temp_2011 <- rast(here("data", "data", "average_annual_sst_2011.tif"))
sea_temp_2012 <- rast(here("data", "data", "average_annual_sst_2012.tif"))

# Read in raster sea bathymetry data
bathymetry <- rast(here("data", "data", "depth.tif"))

# Read in vector economic zones data
economic_zones <- st_read(here("data", "data", "wc_regions_clean.shp"))
```

Check if the datasets have the same coordinate reference systems

```
# Check if bathymetry, sea surface temperature, and economic zones have matching CRS
print(st_crs(bathymetry) == st_crs(sea_temp_2008))
```

```
[1] TRUE
```

```
print(st_crs(bathymetry) == st_crs(economic_zones))
```

```
[1] TRUE
```

Process data

Find the mean SST from 2008-2012 (e.g. create single raster of average SST)

```
# Calculate average sea surface temperature across 2008-2012
sst_mean <- (sea_temp_2008 + sea_temp_2009 + sea_temp_2010 + sea_temp_2011 + sea_temp_2012) / 5
```

Convert average SST from Kelvin to Celsius hint: subtract by 273.15

```
# Covert by subtracting 273.15
sst_mean_c <- sst_mean - 273.15
```

Crop depth raster to match the extent of the SST raster note: the resolutions of the SST and depth data do not match

```
# Crop bathymetry to match SST extent
bathy_crop <- crop(bathymetry, sst_mean_c)
```

Resample the depth data to match the resolution of the SST data using the nearest neighbor approach

```
# Resample the bathymetry raster using method = "near"
bathymetry_resample <- resample(bathy_crop, y = sst_mean_c, method = "near")
```

Check that the depth and SST match in resolution, extent, and coordinate reference system

```
# Check whether the resolution match using the res() function
if(all(res(bathymetry_resample) == res(sst_mean_c))){
  print("resolutions match")
} else {
  print("resolutions do not match")
}
```

```
[1] "resolutions match"
```

```
# Check whether the extents match using the ext() function

if(ext(bathymetry_resample) == ext(sst_mean_c)){
  print("extents match")
} else{
  print("extents do not match")
}
```

```
[1] "extents match"
```

```
# check and transform coordinate reference systems
if(crs(bathymetry_resample) == crs(sst_mean_c)) {
  print("Coordinate reference systems match")
} else{
  warning("Updating coordinate reference systems to match")
  # transform data to match
  bathymetry_resample <- project(bathymetry_resample, crs(sst_mean_c))
}
```

Find suitable locations for Oysters

Oysters research has shown that oysters need the following conditions for optimal growth:

Sea surface temperature: 11-30°C Depth: 0-70 meters below sea level

Reclassify SST and depth data into locations that are suitable for oysters hint: set suitable values to 1 and unsuitable values to 0

```
# Create reclassification matrix for the SST
sst_rcl_mat <- matrix(c(-Inf, 11, 0,
                      11, 30, 1,
                      30, Inf, 0),
                    ncol = 3, byrow = TRUE)

# Use reclassification matrix to reclassify SST raster
sst_reclassified <- classify(sst_mean_c, rcl = sst_rcl_mat)

# Create reclassification matrix for the depth
depth_rcl_mat <- matrix(c(-Inf, -70, 0,
                        -70, 0, 1,
                        0, Inf, 0),
                      ncol = 3, byrow = TRUE)

# Use reclassification matrix to reclassify depth raster
depth_reclassified <- classify(bathymetry_resample, rcl = depth_rcl_mat)
```

Find locations that satisfy both SST and depth conditions

```
# Select only the cells where both conditions are met (multiplication equals 1)
suitable_cells <- sst_reclassified * depth_reclassified
```

Determine the most suitable EEZ

Select suitable cells within West Coast EEZs

```
# Check and transform coordinate reference systems
if(st_crs(economic_zones) == crs(sst_mean_c)) {
  print("Coordinate reference systems match")
} else {
  warning("Updating coordinate reference systems to match")
  # Transform economic_zones to match raster CRS
  economic_zones <- st_transform(economic_zones, crs(sst_mean_c))
}
```

Rasterize the EEZ data

```
# Convert EEZ polygons to raster format using region IDs
ez_rast <- rasterize(economic_zones, suitable_cells, field = "rgn_id")
```

```
# Calculate cell area in km2
cell_area_km2 <- cellSize(suitable_cells, unit = "km")
```

```
# Calculate suitable area (multiply suitable cells by cell area)
suitable_area <- suitable_cells * cell_area_km2
```

```
# Extract suitable area values for each EEZ polygon
eez_areas <- extract(suitable_area, economic_zones, fun = "sum", na.rm = TRUE)
```

```
# Add the results to the economic_zones data
economic_zones$suitable_area_km2 <- eez_areas[, 2]
```

```
# Sort by suitable area (descending)
economic_zones_ranked <- economic_zones[order(-economic_zones$suitable_area_km2), ]
```

Visualize the most suitable EEZ

```
# Convert to data frame and select columns
eez_table <- as.data.frame(economic_zones_ranked)[, c("rgn", "suitable_area_km2")]

# Display ranked EEZ results in formatted table
```

```
kable(eez_table,
      caption = "Top economic zones for Oysters in west coast of US",
      col.names = c("Region", "Suitable Area (km2)"),
      digits = 2, row.names = FALSE) %>%
kable_styling(font_size = 14,
              full_width = FALSE)
```

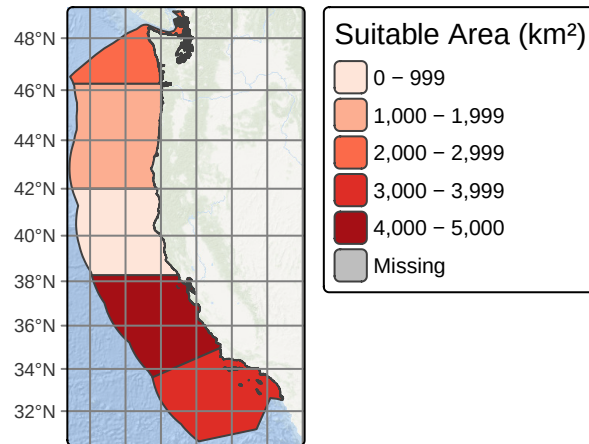
Table 1: Top economic zones for Oysters in west coast of US

Region	Suitable Area (km ²)
Central California	4069.88
Southern California	3757.28
Washington	2378.31
Oregon	1074.27
Northern California	178.03

```
# Plot a map showing the most suitable EEZ
map_suitable_eez <- tm_shape(economic_zones_ranked) +
  tm_polygons(col = "suitable_area_km2",
              palette = "Reds",
              title = "Suitable Area (km2)") +
  tm_title(text = "California EEZs Ranked by Suitable Area\nfor Oyster Aquaculture") +
  tm_graticules() +
  tm_basemap("Esri.OceanBasemap", alpha = 0.7) +
  tm_layout(inner.margins = c(0.01, 0.01, 0.01, 0.1),
            component.autoscale = FALSE)

map_suitable_eez
```

California EEZs Ranked by Suitable Area for Oyster Aquaculture



Make the workflow generalizable for other species by developing a function

Translate the Oyster workflow into a function

```
# Create a function with the arguments: species, min_temp, max_temp, min_depth, max_depth
suitable_eez_fun <- function(species, min_temp, max_temp, min_depth, max_depth){
  # Create reclassification matrix for the SST (make the values generalized)
  sst_rcl_mat <- matrix(c(-Inf, min_temp, 0,
                        min_temp, max_temp, 1,
                        max_temp, Inf, 0),
                      ncol = 3, byrow = TRUE)

  # Use reclassification matrix to reclassify SST raster
  sst_reclassified <- classify(sst_mean_c, rcl = sst_rcl_mat)

  # Create reclassification matrix for the depth (make the values generalized)
  depth_rcl_mat <- matrix(c(-Inf, min_depth, 0,
                          min_depth, max_depth, 1,
                          max_depth, Inf, 0),
                        ncol = 3, byrow = TRUE)

  # Use reclassification matrix to reclassify depth raster
  depth_reclassified <- classify(bathymetry_resample, rcl = depth_rcl_mat)
```

```

# Find locations that satisfy both SST and depth conditions
suitable_cells <- sst_reclassified * depth_reclassified

# Transform economic_zones to match raster CRS (not bathymetry_resample!)
economic_zones_transformed <- st_transform(economic_zones, crs(sst_mean_c))

# Rasterize the EEZ polygons
ez_rast <- rasterize(economic_zones_transformed, suitable_cells, field = "rgn_id")

# Calculate cell area in km²
cell_area_km2 <- cellSize(suitable_cells, unit = "km")

# Calculate suitable area (multiply suitable cells by cell area)
suitable_area <- suitable_cells * cell_area_km2

# Extract suitable area values for each EEZ polygon
eez_areas <- extract(suitable_area, economic_zones_transformed, fun = "sum", na.rm = TRUE)

# Add the results to the economic_zones data
economic_zones_transformed$suitable_area_km2 <- eez_areas[, 2]

# Sort by suitable area (descending)
economic_zones_ranked <- economic_zones_transformed[order(-economic_zones_transformed$suitable_area_km2), ]

# Plot a map showing the most suitable EEZ
map_suitable_eez <- tm_shape(economic_zones_ranked) +
  tm_polygons(col = "suitable_area_km2",
    palette = "Reds",
    title = "Suitable Area (km²)") +
  tm_title(text = paste("California EEZs Ranked by Suitable Area\nfor", species, "Aquaculture"),
    fontface = "bold", fontweight = "bold", fontfamily = "serif", fontcolor = "black",
    size = 12) +
  tm_graticules() +
  tm_basemap("Esri.OceanBasemap", alpha = 0.7) +
  tm_layout(inner.margins = c(0.01, 0.01, 0.01, 0.1),
    component.autoscale = FALSE)

return(map_suitable_eez)
}

```


Choose-your-own species

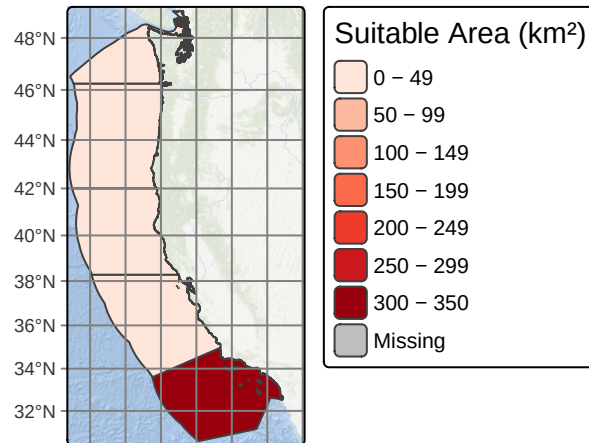
Xiphopenaeus kroyeri (Heller, 1862) Atlantic Seabob shrimp Importance: highly commercial
depth range: 0 - 70 m Temperature: 17°C - 26°C

Apply function for Atlantic Seabob Shrimp

Test the function on the new range of temperature and depths

```
# Apply the suitable EEZ function for Atlantic seabob shrimp  
suitable_eez_fun("Atlantic Seabob Shrimp", 17, 26, -70, 0)
```

California EEZs Ranked by Suitable Area for Atlantic Seabob Shrimp Aquaculture



References

NOAA Coral Reef Watch Daily 5km Satellite Coral Bleaching Heat Stress SST Anomaly Product (Version 3.1). (2018). NOAA.gov. https://coralreefwatch.noaa.gov/product/5km/index_5km_ssta.php

Gridded Bathymetry Data. (2025). GEBCO. <https://www.gebco.net/data-products/gridded-bathymetry-data#area>

Marine Regions. (2025). Marineregions.org. <https://www.marineregions.org/eez.php>

Commercially Important Species Occurring in United States (contiguous states). (2025). Sealifebase.ca. https://www.sealifebase.ca/country/CountryChecklist.php?c_code=840&vhabitat=commercial